

Competition Rules

❖ Brake System Plausibility Device

Functionality:

The BSPD is a standalone safety circuit connected directly to the Shutdown Circuit (SDC). It monitors the relationship between braking and motor power. If the driver applies heavy braking while the motors are still receiving significant power for longer than the allowed time, the BSPD opens the SDC and shuts down the tractive system.

The BSPD must stay triggered until the fault is removed for a required time or the low-voltage master switch is power-cycled. It must be a **non-programmable hardware circuit**, meaning it uses analog components/relays instead of a microcontroller.

Importance:

The BSPD protects against a dangerous **"braking while accelerating"** situation, such as a stuck throttle or electrical fault. Without it, the car could lose control, damage the drivetrain, or create a fire risk.

❖ Insulation Monitoring Device

Functionality:

The IMD continuously checks the insulation resistance between the high-voltage system and the vehicle chassis. If insulation becomes too low, meaning the high-voltage system could leak current to the chassis, the IMD opens the Shutdown Circuit.

Importance:

A damaged insulation layer can make the vehicle chassis electrically dangerous and expose drivers and team members to electric shock. The IMD provides automatic protection by detecting insulation failures and disconnecting the high-voltage system.

❖ Accumulator Isolation Relay

Functionality:

AIRs are high-current relays that connect or disconnect the accumulator (battery pack) from the tractive system. At least two AIRs are required, one on the positive rail and one on the negative rail, to fully isolate the battery.

They are controlled by safety systems such as the Shutdown Circuit, IMD, and BSPD. Their actual state is also monitored by the TSAL system to determine whether the vehicle is safe to touch.

Importance:

The AIRs are the main physical disconnect between the high-voltage battery and the rest of the vehicle. When a safety fault occurs, opening the AIRs removes dangerous high voltage from the system.

❖ Brake over travel switch

Functionality:

The BOTS is a mechanical switch connected to the Shutdown Circuit. It detects excessive brake pedal movement caused by a failure in one of the brake circuits.

If the brake pedal travels further than normal, the BOTS opens the Shutdown Circuit and shuts down the vehicle. The driver cannot reset it, and repeated activation must not automatically close the circuit again.

Importance:

If one brake circuit fails, the pedal may travel too far before braking happens. The BOTS detects this failure immediately and prevents the driver from continuing with unsafe brakes.

❖ Inertia switch

Functionality:

The inertia switch is a mechanical safety device connected to the Shutdown Circuit. It activates when it detects a sudden deceleration, such as during a crash, and disconnects the tractive system.

Importance:

After an accident, the vehicle must shut down automatically to prevent further hazards such as electrical arcing, unwanted motor movement, or fire. The inertia switch provides this protection without requiring driver action.

❖ Tractive system active light

Functionality:

The TSAL shows the status of the high-voltage system using two independent lights:

- Red light: Flashes when the tractive system is active (high voltage present above the safety limit).
- Green light: Turns on when the tractive system is safely disconnected, meaning the AIRs are open, pre-charge is complete, and dangerous voltage is not present.

The two circuits must work independently and must be made using **hard-wired electronics**, not software.

Importance:

The TSAL provides a clear external indication of whether the high-voltage system is dangerous or safe to work on. It prevents mistakes by drivers, engineers, and marshals by giving reliable visual safety signal.